



جامعة أم القرى
UMM AL-QURA UNIVERSITY

Kingdom of Saudi Arabia

Ministry of Education

Umm Al-Qura University

College of Engineering and Computers at Al-Qunfudah

Bachelor of Science in Computer Science

Group Project: Deploying a Static HTML Website Using Amazon S3

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Prepared For:

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Cloud Computing Project: Deploying a Static HTML Website Using Amazon S3

1. Personal Information

AYAK Public Website URL: <https://ayak-website-bucket.s3website.eu-north-1.amazonaws.com/>

d1ymgw7t5nrgum.cloudfront.net

ID	Name	Role
444*****	Eng. Khaled Mahmoud Sulaimani	Team Leader / AWS Specialist/ Web Developer/ Documentation Specialist

2. Project Description

This website serves as the digital front page for "AYAK", a tech startup specializing in providing smart, innovative solutions in the Internet of Things (IoT) and embedded systems. Developed using HTML, CSS, and JavaScript, the site is designed to professionally showcase the platform's vision, IoT hardware, and project team through a modern and interactive user experience.

3. Cloud Implementation Steps

The website files were successfully hosted on a cloud environment using Amazon S3. Below is the documentation of the implementation process:

Step 1: Creating the Storage Space (S3 Bucket) A new bucket was created under the AWS Free Tier with a globally unique name (`ayak-website-bucket`) to serve as the main container for the website files.

Object Ownership

eu-north-1.console.aws.amazon.com/s3/bucket/create/?ref=

[Alt+S] Ask Amazon Q

Europe (Stockholm)

Create bucket

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through new public bucket or access point policies**

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Turning off Block all public access might result in this bucket and the objects within becoming public

AWS recommends that you turn on Block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

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Step 2: Security Settings & Public Access To make the website accessible to visitors, the "Block all public access" setting was disabled. A Bucket Policy written in JSON was then attached to grant read-only permissions (`s3:GetObject`) to the public.

Step 3: Enabling Static Website Hosting The static website hosting feature was enabled from the bucket's properties. The `index.html` file was specified as the default index document to load when visiting the endpoint.

[Alt+S] Ask Amazon Q Europe (Stockholm)

website-bucket > Edit static website hosting

Edit static website hosting [info](#)

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting
☐ Disable
☒ Enable

Hosting type
☒ Host a static website
 Use the bucket endpoint as the web address. [Learn more](#)
☐ Redirect requests for an object
 Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document
Specify the home or default page of the website.

Error document - optional
This is returned when an error occurs.

Redirection rules - optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. [Learn more](#)

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Step 4: Uploading Website Files All project files, including HTML, CSS, and image assets, were successfully uploaded to the S3 bucket.

Upload succeeded
For more information, see the Files and folders table.

Summary

Destination: s3://zyak-website-bucket

Succeeded: 6 files, 1.3 MB (100.00%)

Failed: 0 files, 0 B (0%)

Files and folders Configuration

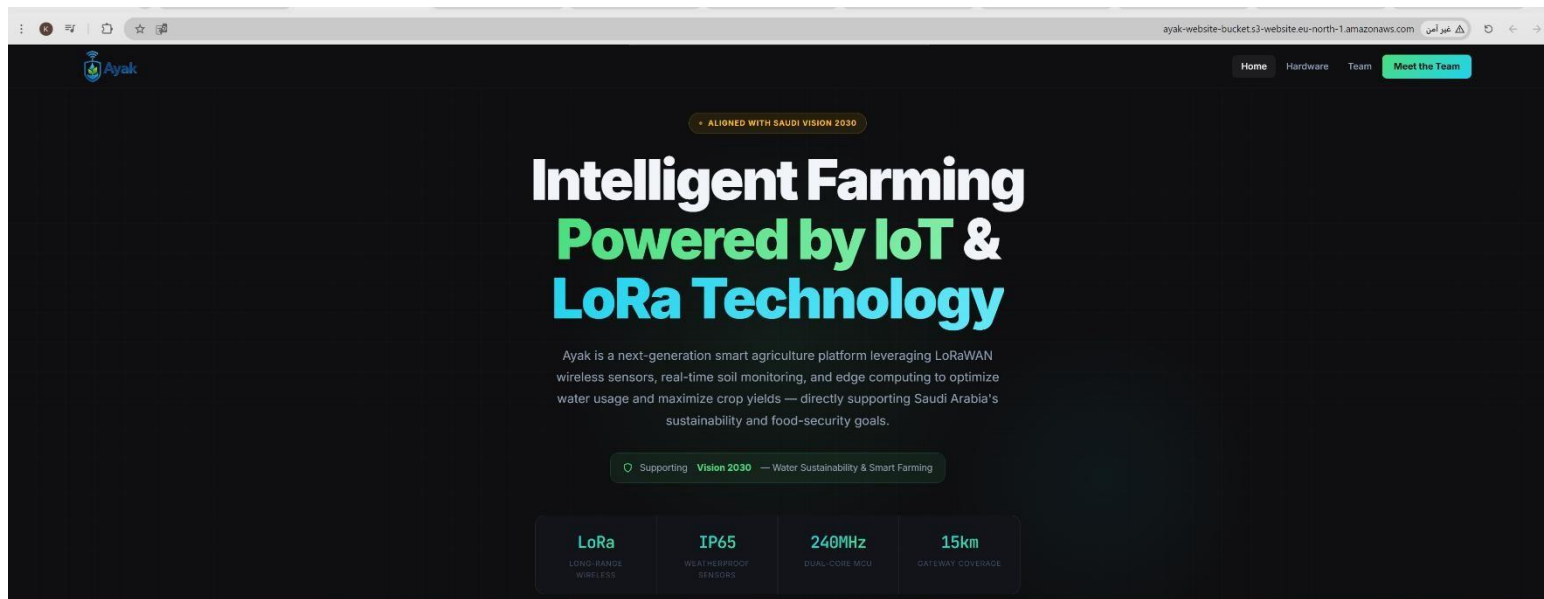
Files and folders (6 total, 1.3 MB)

Find by name

Name	Folder	Type	Size	Status	Error
zyak_logo_1_shield_wif_transparen...	-	image/png	460.1 KB	✓ Succeeded	-
dfrobot_gateway_eu868.png	-	image/png	225.9 KB	✓ Succeeded	-
Gravity IP65 Capacitive Soil Moistur...	-	image/png	264.6 KB	✓ Succeeded	-
heltec_esp32_lora_v3.png	-	image/png	260.6 KB	✓ Succeeded	-
index.html	-	text/html	36.7 KB	✓ Succeeded	-
team.html	-	text/html	32.8 KB	✓ Succeeded	-

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Step 5: Final Result & Deployment The public endpoint URL was tested via a web browser. The static website loaded successfully and is now accessible to the public.



4. Security Awareness & S3 Bucket Policies

As part of understanding the shared responsibility model in cloud computing, we analyzed the security implications of hosting a static website on Amazon S3:

- **S3 Bucket Policies:** These are JSON-based security rules used to manage access permissions at the bucket level. For this project, we configured the policy to allow anonymous users (the public) to read objects ("Action": "s3:GetObject") without granting them any permissions to modify, upload, or delete data, which is the standard practice for static web hosting.
- **Security Risks of Enabling Public Access:** Disabling the "Block Public Access" feature carries inherent risks. If sensitive data (such as passwords, API keys, personal user data, or internal company documents) is accidentally uploaded to this public bucket, it will be exposed and accessible to anyone on the internet. Furthermore, unrestricted public access can make the bucket vulnerable to malicious scraping or bot traffic, which can consume significant bandwidth and result in unexpected financial charges on the AWS account.